

## Experiment No. 19

### *Write a Prolog Program for PLANETS DB*

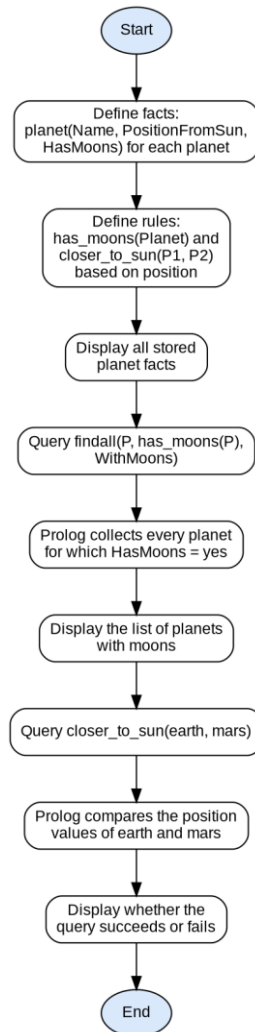
#### **Aim**

To write a Prolog program that stores facts about the planets of the solar system (their position from the sun and whether they have moons), and answers queries about them.

#### **Algorithm**

1. Start.
2. Define facts of the form `planet(Name, PositionFromSun, HasMoons)` for each of the eight planets.
3. Define the rule `has_moons(Planet) :- planet(Planet, _, yes)`.
4. Define the rule `closer_to_sun(P1, P2) :- planet(P1, Pos1, _), planet(P2, Pos2, _), Pos1 < Pos2`.
5. Display all the stored planet facts.
6. Query `findall(P, has_moons(P), WithMoons)` to collect every planet that has moons.
7. Display the resulting list of planets with moons.
8. Query `closer_to_sun(earth, mars)` to check whether Earth is closer to the Sun than Mars.
9. Prolog compares the position values of the two planets and succeeds or fails accordingly.
10. Display the result of that query.
11. Stop.

#### **Flowchart**



## Program

```

planet(mercury, 1, no).
planet(venus, 2, no).
planet(earth, 3, yes).
planet(mars, 4, yes).
planet(jupiter, 5, yes).
planet(saturn, 6, yes).
planet(uranus, 7, yes).
planet(neptune, 8, yes).

```

```

has_moons(Planet) :- planet(Planet, _, yes).

```

```

closer_to_sun(P1, P2) :-
    planet(P1, Pos1, _),

```

```

planet(P2, Pos2, _),
Pos1 < Pos2.

main :-
    format("PLANETS Database~n"),
    format("-----~n"),
    forall(planet(P, Pos, Moons),
        format("~w~t~12|position ~w~t~25|has moons:
~w~n", [P, Pos, Moons])),
    format("~nQuery: planets that have moons~n"),
    findall(P2, has_moons(P2), WithMoons),
    format("Result: ~w~n", [WithMoons]),
    format("~nQuery: is earth closer to the sun than
mars?~n"),
    ( closer_to_sun(earth, mars)
    -> format("Result: yes, earth is closer to the sun than
mars~n")
    ; format("Result: no~n")
    ).

```

## Result

The program was executed successfully and the output was verified as correct.

?- main.

PLANETS Database

```

-----
mercury      position 1   has moons: no
venus        position 2   has moons: no
earth        position 3   has moons: yes
mars         position 4   has moons: yes
jupiter      position 5   has moons: yes
saturn        position 6   has moons: yes
uranus        position 7   has moons: yes
neptune       position 8   has moons: yes

```

Query: planets that have moons

Result: [earth,mars,jupiter,saturn,uranus,neptune]

Query: is earth closer to the sun than mars?

Result: yes, earth is closer to the sun than mars

true.

## Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal:

The screenshot shows the Visual Studio Code interface with the file `planets.pl` open in the editor. The Explorer sidebar on the left shows the project structure under `AI-Lab`, including files like `sum_n.pl`, `name_dob.pl`, `student_teacher.pl`, `planets.pl` (selected), and `hanoi.pl`.

The `planets.pl` file contains the following Prolog code:

```
1 planet(mercury, 1, no).
2 planet(venus, 2, no).
3 planet(earth, 3, yes).
4 planet(mars, 4, yes).
5 planet(jupiter, 5, yes).
6 planet(saturn, 6, yes).
7 planet(uranus, 7, yes).
8 planet(neptune, 8, yes).
9
10 has_moons(Planet) :- planet(Planet, _, yes).
11
12 closer_to_sun(P1, P2) :-
13     planet(P1, Pos1, _),
14     planet(P2, Pos2, _),
15     Pos1 < Pos2.
16
17 main :-
18     format("PLANETS Database-n"),
19     format("-----n"),
20     forall(planet(P, Pos, Moons),
21         format("~w-t-12|position ~w-t-25|has moons: ~w-n", [P, Pos, Moons])),
22     format("~nQuery: planets that have moons-n"),
23     findall(P2, has_moons(P2), WithMoons),
24     format("Result: ~w-n", [WithMoons]),
25     ! (code continues — see Program section).
```

The integrated terminal at the bottom shows the execution of the program using the `swipl` command. The output is as follows:

```
PS C:/Users/Athavan/AI-Lab> swipl planets.pl
Welcome to SWI-Prolog (threaded, 64 bits, version 9.0.4)

?- main.
PLANETS Database
-----
mercury    position 1    has moons: no
venus      position 2    has moons: no
earth      position 3    has moons: yes
mars       position 4    has moons: yes
jupiter    position 5    has moons: yes
saturn     position 6    has moons: yes
uranus     position 7    has moons: yes
neptune    position 8    has moons: yes
Query: planets that have moons
Result: [earth,mars,jupiter,saturn,uranus,neptune]
Query: is earth closer to the sun than mars?
Result: yes, earth is closer to the sun than mars
true.
?-
```

The status bar at the bottom indicates the current position is at line 1, column 1, and the file is encoded in UTF-8 with LF line endings and 4 spaces per tab.